Technical specification template: UPS Batteries

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A5.
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Document control

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1. Scope

This template defines the technical requirements for ups batteries for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEC 60896 series (stationary lead-acid), if lead-acid is selected
- IEC 62619 (secondary lithium cells and batteries, safety), if lithium-ion is selected
- IEC 62485-2 (safety requirements for stationary batteries)
- IEEE 485 (sizing lead-acid batteries) as a method reference
- Local fire and ventilation regulations [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit study pending utility, transformer and generator data]	NOT CALCULATED

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Autonomy to be provided	10 min at the critical load	PRELIMINARY_ASSUMPTION
Energy at the UPS output	333 kWh (arithmetic, not a battery size)	Preliminary
DC bus voltage, cell count, strings	[To be determined by the project]	Pending manufacturer constant-power discharge data
Chemistry and design life	[Project to select]	Project decision
End-of-life capacity, temperature and design margin factors	[To be determined by the project]	Pending manufacturer data

5. Technical requirements

5.1 The battery shall deliver the stated autonomy at the stated constant power to the end-of-discharge voltage at end of life and at the minimum room temperature.

5.2 The manufacturer shall provide the constant-power discharge table (power per cell against time to the end voltage) for the offered cell.

5.3 The manufacturer shall state ageing, temperature correction and expected life and shall declare the capacity at the design discharge rate.

5.4 Battery management, monitoring, protection, racks, seismic requirements and ventilation or gas management: [project to define].

5.5 Thermal runaway and fire safety measures: [project to define with the authority].

6. Manufacturer data required

The tenderer shall supply the datasheets and values below with its offer, each with the document name and revision. A value without a source document is not accepted. These are also tracked in the manufacturer-data register and are imported into the calculations through the data override mechanism.

Parameter id	Parameter	Datasheet	Current status
battery.dc_bus_voltage_v	DC bus voltage	Battery data sheet with the constant-power discharge table (W per cell to the end voltage at the autonomy time), temperature and ageing data	MISSING
battery.inverter_efficiency_pct	Inverter efficiency on battery	Battery data sheet with the constant-power discharge table (W per cell to the end voltage at the autonomy time), temperature and ageing data	MISSING
battery.end_of_life_cap	End-of-life capacity	Battery data sheet with	MISSING

acity_factor	factor	the constant-power discharge table (W per cell to the end voltage at the autonomy time), temperature and ageing data	
battery.temperature_correction_factor	Temperature correction factor	Battery data sheet with the constant-power discharge table (W per cell to the end voltage at the autonomy time), temperature and ageing data	MISSING
battery.constant_power_discharge_data	Constant-power discharge data reference	Battery data sheet with the constant-power discharge table (W per cell to the end voltage at the autonomy time), temperature and ageing data	MISSING

7. Inspection and quality requirements

7.1 Review of the cell datasheet, rack drawings and safety documentation.

7.2 Inspection of delivery condition, cell voltages and torque.

7.3 Routine tests on every unit as required by the applicable standard and this specification.

7.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

7.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

7.6 Calibration certificates for test instruments.

8. Factory acceptance test (FAT) requirements

8.1 Cell and module routine tests per the standard.

8.2 Capacity and discharge tests on a sample [project to define].

8.3 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9. Commissioning requirements

9.1 Installation checks, polarity, torque, insulation resistance.

9.2 Initial charge per the manufacturer.

9.3 Capacity or autonomy test with the UPS at the design load; records retained as the baseline.

9.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

10. Submittals and documentation

- Datasheets, drawings and type-test reports.
- Inspection and test plan, FAT procedure and report.
- Installation, operation and maintenance manuals.
- Spare parts and special tools lists.
- Warranty terms [project to define].

11. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- This template has not been reviewed by an independent engineer.